



SOMAIYA
VIDYAVIHAR UNIVERSITY

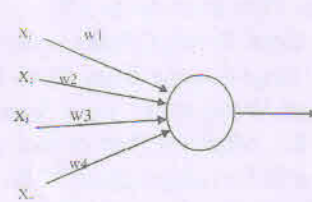
Semester: August 2022 – December 2022 (Jan-2023)		
Maximum Marks: 100	Examination: ESE Examination (Re-Exam)	Duration: 3 hrs.
Programme code: 116U01	Class: TY	Semester: V (SVU 2020)
Programme: BTech Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMPUTERS
Course Code: 116U01C501	Name of the Course: Software Engineering	
Instructions: 1) Draw neat diagrams 2) Assume suitable data if necessary		

Question No.		Max. Marks
Q1 (a)	Draw the Use Case diagram and activity diagram for Online Job Portal System in which an administrator generates vacancy details and gets them approved and uploads them on the website. The job seeker first registers himself to see the vacancy and uploads his CV which will be notified to the administrator. OR Develop a class diagram for a doctor who records the details of the patient after each visit and updates it with recent details and saves it. An attendee is given access to see the details for printing if necessary and see the details of the prescribed drugs.	10
Q1 (b)	Explain the Agile process with its advantages. Explain any one Agile process model?	10
Q2 (a)	Create a Software Requirement Specification for an online shopping system. SRS should include the following: (a) Product perspective (b) Scope and objective (c) Functional requirements (d) Non-functional requirements OR Draw Sequence and Collaboration diagrams for an online food ordering system.	10
Q2 (b)	List different metrics used for software measurement. Explain the function point-based estimation technique in detail.	10
Q3 (a)	Explain what is cyclomatic complexity and the different methods to calculate it. Find the cyclomatic complexity of the following code: <pre>int x, y, power; float z; input (x, y); if (y<0) power = -y;</pre>	10

	<pre> else power = y; z = 1; while (power! = 0) { z = z * x; power = power - 1; if (y<0) z = 1/z; output (z); end </pre>	
Q3 (b)	Assume that the library Management system is deployed in a client-server architecture. Explain the various components and their deployment using a diagram.	10
Q4 (a)	What is the significance of User interface in system development? Draw a graphical user interface for an online shopping system. OR Explain cohesion and coupling. What are the benefits of high cohesion and low coupling?	10
Q4 (b)	Prepare a risk identification checklist and RMMM plan for creating a UID with biometrics (Unique identification number) for a highly populated country.	10
Q5 (a)	How will Map Object Model to Database Schema? Explain with a suitable example. OR Explain steps in version and change control.	10
Q5 (b)	What is FTR? Explain the review guidelines considered during FTR.	10



Semester: August 2022 – December 2022		
Maximum Marks:100	Examination: ESE Examination(<i>Re-exam</i>) Duration: 3 Hours	
Programme code: 01	Class: TY	Semester: V (SVU 2020)
Programme: B Tech in Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP
Course Code: 116U01E514	Name of the Course: Soft Computing	
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary		

Question No.		Max. Marks
Q1	Solve any Four (i) Discuss Biological neuron and its working (ii) Compare Hard computing and soft computing (iii) Describe Delta Learning (iv) Discuss the concept of Linear Separability. (v) Describe Support Vector Machine (SVM) (vi) Consider two fuzzy sets $A = \{0.4/x_1 + 0.3/x_2 + 0.9/x_3 + 1/x_4\}$ and $B = \{0.1/x_1 + 0.2/x_2 + 0.7/x_3 + 0.9/x_4\}$, Find fuzzy set C which is the union between A and B. Find the Complement of set B.	20
Q2 (a)	(i) Write a short note on Learning Parameters (ii) Assume the network given in the figure below with the initial weights w_1 and with continuous bipolar activation function $f(net) = \text{sgn}(net)$ Input vector is given as $X_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$  The weight and three input vectors are as below $w_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix} \quad X_1 = \begin{bmatrix} 0 \\ -1 \\ 1.5 \\ 0 \end{bmatrix} \quad X_2 = \begin{bmatrix} 0 \\ -0.5 \\ 1 \\ 2 \end{bmatrix} \quad X_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$ Update the weights. Show only one iteration updates. using Hebbian Learning Rule.	10

Q2 (b)	List and discuss in short, neural network learning rules and give their comparison with respect to weight adjustment equation, initial weights, neuron characteristics and learning type (supervised and unsupervised). OR Elaborate the Error Back Propagation Training Algorithm (EBPTA) in detail with a suitable example.	10																						
Q3 (a)	Elaborate Bidirectional Associative Memory (BAM)	10																						
Q3 (b)	Write a note on Kohonen Self Organizing Feature Map OR Explain Hopfield Network with a suitable example	10																						
Q4	(i) Explain α cut set and strong α cut set of a fuzzy set. Find the strong α cut and α cut for the fuzzy set A given below, for $\alpha = 0.2$, and $\alpha = 0.6$. Fuzzy set, $A = \{(0, 0.1), (3, 0.3), (5, 0.4), (6, 0.3), (8, 0.9), (9, 0.4)\}$ (α cut set is also called as α level set.) (ii) List defuzzification techniques and explain any one in brief. OR Apply the fuzzification model to the given data of the Washing machine where the input variables are Dirt and Grease present in the cloths to be washed. The linguistic variables for Dirt, are Small Dirt (SD), Medium Dirt (MD) and Large Dirt (LD). The linguistic variables for Grease, are Small Grease (SG), Medium Grease (MG), Large Grease (LG). Find out the wash time required if Dirt = 80 and Grease = 50. The linguistic variables for the output wash time are Very Small (VS), Small (S), Medium (M), large (L), Very Large (VL) The triangular membership functions for the input and output variables are as below: The input Small Dirt value ranges for 0 to 40, Medium Dirt from 0 to 100 and Large Dirt from 40 to 100 The input Small Grease value ranges from 0 to 40, Large Grease from 0 to 100 and Large Grease from 40 to 100 Output Wash time, Very Small (VS) in the range 0 to 10, Small (S) in the range 0 to 25, Medium (M) in the range 10 to 40, Large (L) in the 25 to 60 and Very Large (VL) ranges from 40 to 60 minutes. Find the exact wash time required after defuzzification by using Mean of maximum method. The Decision rule table is as below <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td colspan="2"></td> <th colspan="3">Grease</th> </tr> <tr> <td rowspan="4" style="writing-mode: vertical-rl; transform: rotate(180deg);">Dirt</td> <td></td> <td>SG</td> <td>MG</td> <td>LG</td> </tr> <tr> <td>SD</td> <td>VS</td> <td>M</td> <td>L</td> </tr> <tr> <td>MD</td> <td>S</td> <td>M</td> <td>L</td> </tr> <tr> <td>LD</td> <td>M</td> <td>L</td> <td>VL</td> </tr> </table>			Grease			Dirt		SG	MG	LG	SD	VS	M	L	MD	S	M	L	LD	M	L	VL	20
		Grease																						
Dirt		SG	MG	LG																				
	SD	VS	M	L																				
	MD	S	M	L																				
	LD	M	L	VL																				
Q5	Write notes / Short question type on any four (i) Describe operations of Fuzzy sets with examples. (ii) Discuss Fuzzy membership functions (iii) Write a short note on 'Outstar Learning' (iv) Discuss Adaptive resonance theory (ART) (v) Explain the possibility theory of fuzzy sets. (vi) Write a short note on Boltzman Machine	20																						



Semester: August 2022 – December 2022 - Jan-2023		
Maximum Marks:100	Examination: ESE Examination - (Re-exam) Duration: 3 Hours	
Programme code: 01	Class: TY	Semester: V
Programme: B Tech Computer Engineering	(SVU 2020)	
Name of the Constituent College:	Name of the department:	
K. J. Somaiya College of Engineering	COMP	
Course Code: 116U01C502	Name of the Course: Computer Networks	
Instructions: 1)Draw neat diagrams. 2)Assume suitable data if necessary		

Question No.		Max. Marks
Q1 (a)	Draw and explain the OSI Reference Model with services and functions of each layer. OR Discuss design issues in OSI Layers	10 10
Q1 (b)	A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted? OR Explain CSMA/CD in detail with its frame format.	10 10
Q2 (a)	The following is a dump of a UDP header in hexadecimal form: 06 32 00 0D 00 1C E2 17 What is the (a) Source port number (b) Destination port number (c) Total length of the UDP (d) Length of the data (e) Considering that an IP frame can have a maximum total length of 65535 bytes, what is the maximum length of the data in a UDP frame?	10
Q2 (b)	What is Domain Name System? How does it work? Explain the resolution process. OR Explain HTTP Protocol with its features and working.	10 10
Q3 (a)	A slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces: (a) 1000 frames per second (b) 500 frames per second (c) 250 frames per second.	10
Q3 (b)	An address in a block is given as 201.10.6.54. Find the number of addresses in the block, the first address, and the last address. <i>Assume classful addressing.</i> OR Explain IPv6 Packet Format in detail with neat diagram.	10 10
Q4 (a)	Explain TCP segment header format in detail. OR	10

	Explain Open Loop and Closed Loop Congestion Control in TCP.	10
Q4 (b)	Explain OSPF protocol in detail with its working and message format.	10
Q5 (a)	Explain SMTP Protocol in detail with its components and working.	10
Q5 (b)	Discuss the different networking devices used for internetworking.	10
	OR	
	Draw and explain different networking topologies.	10



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Semester: August 2022 – December 2022 – Jan-2023		
Maximum Marks: 100	Examination: ESE Examination – Re-Exam Duration:3 Hrs.	
Programme code: 01	Class: TY	Semester: V (SVU 2020)
Programme: Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116U01C503	Name of the Course: Operating System	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What is operating system? List the objectives and functions of operating system	5
ii)	Differentiate between monolithic and structured layer of OS	5
iii)	Draw and explain 7-state diagram of life cycle of a process	5
iv)	Explain semaphores and its different types .	5
v)	Explain hierarchical paging	5
vi)	What is segmentation? State its usage.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Suppose that a disk drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at cylinder 2150 and the previous request was at cylinder 1805. The queue of pending requests in FIFO order is 2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681. Starting from the current head position, what is the total distance(in cylinders) that the disk arm moves to satisfy all pending requests for the following Disk scheduling algorithms (i) SCAN (ii) C-LOOK	5
ii)	Explain any two I/O Buffering techniques	5
OR		
Q2 A	What are the various allocation methods with reference to file systems	10
Q 2 B	Solve any One	10
i)	What are the five major categories of system calls? Explain giving one example from each	10
ii)	Differentiate between Network OS and Distributed OS	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	What are the requirements of mutual exclusion? Give software approaches to achieve mutual exclusion.	10
ii)	What is deadlock? Explain various deadlock prevention techniques.	10
iii)	What is critical section problem? Explain different ways to solve critical section problem	10

Que. No.	Question	Max. Marks																								
Q4	Solve any Two	20																								
i)	Consider the following set of Processes with the length of CPU burst given in ms <table><tr><th>Process</th><th>Burst time</th><th>Priority</th><th>Arrival time</th></tr><tr><td>P1</td><td>27</td><td>5</td><td>0</td></tr><tr><td>P2</td><td>6</td><td>3</td><td>3</td></tr><tr><td>P3</td><td>7</td><td>1</td><td>5</td></tr><tr><td>P4</td><td>10</td><td>2</td><td>10</td></tr><tr><td>P5</td><td>15</td><td>4</td><td>13</td></tr></table> Assume smaller number implies higher priority. (i) Draw the Gantt chart for FCFS, SJF, Priority (Pre-emptive) and Round Robin (Quantum = 2). (ii) Which algorithm results in maximum average waiting time? (iii) Calculate average turnaround time for each of the above algorithms	Process	Burst time	Priority	Arrival time	P1	27	5	0	P2	6	3	3	P3	7	1	5	P4	10	2	10	P5	15	4	13	10
Process	Burst time	Priority	Arrival time																							
P1	27	5	0																							
P2	6	3	3																							
P3	7	1	5																							
P4	10	2	10																							
P5	15	4	13																							
ii)	Explain Multithreaded model of a process..	10																								
iii)	Explain multilevel Queue scheduling	10																								

Que. No.	Question	Max. Marks
Q5	Answer any four	20
i)	Write a short note on File organization.	5
ii)	Explain the role of PCB in Process management. Why are two modes of execution needed	5
iii)	Discuss the semaphore solution for Reader/ Writer Problem.	5
iv)	Write short note on Virtual memory.	5
v)	Explain Demand paging	5
vi)	Discuss the monitor solution for dining philosopher problem	5